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A Universal Foundation Model bridging Molecular Discovery and Industrial Scale-Up

MICRO TO MACRO

One continuous mathematical space for physical reasoning.

UNIPHY

A 10-billion parameter foundation model that computes physics in a single latent space.



Micro / Quantum

Electron bonding and molecular affinities.



Meso / Fluids

Viscosity and non-Newtonian rheology.



Macro / Factory

Impeller shear, mixing time, and thermal dynamics in 10,000L vats.

The same model designs the molecule and the factory process simultaneously.

PHYSICAL MANUFACTURING

What cloud computing did for software, UNIPHY does for factories.



Virtualizing Innovation

Transitioning physical Design of Experiments (DoE) entirely in silico. We simulate 10,000 failures computationally so the wet-lab only tests the guaranteed winners.



The Engine for Autonomous Factories

Shodh AI is not just a chatbot. It is the core "Physics Brain" that future AI agents will use to autonomously monitor, control, and optimize chemical and biological factories worldwide.



The Goal

Moving India from a hub of contract execution to the global epicenter of high-value scientific IP and process intelligence.

CURRENT SECTORS & COMMERCIAL TRACTION

Emergent properties proven across overlapping industries.

PILOT PARTNERS

National Resilience, Axella Biotech, A123, Aarti Industries, Jubilant Ingrevia, and Biosys.



Target Verticals

Specialty Chemicals, Pharmaceuticals, Petrochemicals, and Battery Materials — all governed by the same underlying fluid and thermodynamic laws.



Commercial Proof

We have already executed computational pilots with industry leaders like National Resilience (US), Axella Biotech (US), A123 (US), Aarti Industries (Chem), Jubilant Ingrevia, and Biosys (Chem and Bio CRO).



The Output

For these partners, the model successfully synthesized novel molecular routes and predicted complex scale-up physics, validating our Sim-to-Real translation.

Beyond standard retrosynthesis. It uses first-principles thermodynamics to invent novel, cheaper catalytic routes that bypass existing patents and optimize lab yields.

SHODH SYNTH / THE PROCESS

**SHODH
SYNTH**

PROCESS INTELLIGENCE



The ultimate Bioreactor Digital Twin. Ingests bench-scale chemistry to predict thermal runaways and non-Newtonian fluid behaviors in massive Continuous Stirred-Tank Reactors (CSTRs).

SHODH SCALE / THE FACTORY

**SHODH
SCALE**

FACTORY INTELLIGENCE



PLATFORM VALIDATION

Small Molecule Case Study

To stress-test our core physics engine, we successfully executed a fully autonomous Hit-to-Lead pipeline.



Zero-Shot Target Discovery

We fed the model complex oncology targets (e.g., p53, KRAS). Instead of relying on historical templates, it utilized first-principles thermodynamics to natively generate highly potent, structurally novel hits.



Multi-Parametric Optimization (MPO)

The model autonomously mutated scaffolds to avoid "Molecular Obesity." It successfully balanced sub-nanomolar binding affinity with strict clinical ADMET safety proxies: Caco-2 permeability, CYP450 clearance, and hERG safety gaps.



The Output (Ready for Wet-Lab)

The system generated a computationally de-risked, 18,000-molecule proprietary IP catalog. The top candidates include complete, commercial retrosynthesis routes and are currently staged for physical SPR validation.

ADVANCED MODALITIES

Advanced Modalities: Large Molecules & Biomanufacturing

BIOMANUFACTURING

Optimize oxygen transfer without destroying biological yield.



Lipid Nanoparticles (LNPs)

We model the thermodynamic self-assembly and phys-chem stability of LNPs for mRNA and biologic delivery.



Peptides & PROTACs

Designing highly flexible, "chameleonic" structures that break Lipinski's rules but maintain cell permeability.



The Scale-Up of Living Cells

Shodh SCALE directly addresses the Biomanufacturing bottleneck. We simulate the exact fluid shear-stress generated by impeller blades against the rupture limits of living mammalian cells—optimizing oxygen transfer without destroying the biological yield.